

to this effect. This is because a directory should be opened before performing these operations, and directory opening is controlled by the *execute* permission.

6. Provide *read* and *execute* permissions will allow the listing of the directory's contents and the opening of the directory, but creation of new contents will not be allowed.

Adding and managing users without user management commands

Method 1: The GUI interface can be used to manage users in CentOS. In your Linux box:

1. Click on *System*, which appears on the top bar of your desktop.
2. Click on *Administration*.
3. Click on *Users and Groups*.
4. The *User Manager Interface* will be displayed.
5. Here, you can use the *Add User* option to create new users with specific configurations.
6. To modify the configurations of existing users, click on the name of a user and click *Properties*. Here you can update the basic user settings, account settings, password aging and group membership management (see Figure 4).
7. To delete a user, simply select a user from the list and click the *Delete* button.
8. Similarly, groups can be managed from the 'Groups' tab.



Note: *User Manager* can also be started from the terminal using the following command:

```
system-config-users
```

Method 2: Make all the configurations and entries manually. Then add a new user named *test2*, whose primary group will be *test1* and home directory will be stored in */mnt/test2home* (there will be no group with the name *test2*).

1. As root user, edit */etc/passwd* and make the following entry:

```
test2 : x : 503 : 502 : : /mnt/test2home : /bin/bash
```

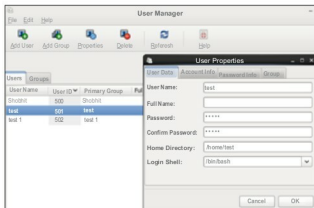


Figure 4: The CentOS user manager GUI interface

2. Create the home directory:

```
mkdir /mnt/test2home
```

3. Edit */etc/group* and make *test2* a member of the group *test1*, by modifying the entry of the group *test1* as follows:

```
test1 : x : 502 : test2
```

4. Now, generate a SHA-512 hashed password for the *test2* user. This password will be stored in the */etc/shadow* file:

```
\sbin/grub-crypt -s sha-512
```

5. Copy the generated hashed password.
6. Now, open the */etc/shadow* file and make the following entry:

```
test2 : <copy-hashed-password-here> : 16498 : 0 : 99999 : 7 : :
```

The values may be modified as required for password aging.

7. Now, give the new users permission on their home directory as follows:

```
chmod 700 /mnt/test2home
```

8. Make *test2* as the user owner and *test1* as the group owner of this home directory:

```
chown test2:test1 /mnt/test2home
```

9. Now, copy the bash shell configuration files from */etc/skel* to */mnt/test2home*:

```
cp /etc/skel/.bashrc /mnt/test2home
cp /etc/skel/.bash_profile /mnt/test2home
```

10. Log in as *test2* using the switch user command:

```
su - test2
```

11. Also, log out from root through the GUI and try to log in with *test2* from the GUI login window. The new user should work fine.

Well implemented permissions can secure your Linux machine to a great extent. Security implementation skills act as the basic building blocks of systems administration and must be worked on continuously. 

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